

Roll No. _____

26/8150**B.C.A. (Fifth Semester)****Examination, 2026****Third Paper****(Computer Networks)***Time : Two Hours]**[Maximum Marks : 75***Note :** Attempt **all** sections as per instructions.**Section-A****(Very Short Answer Type Questions)****Note :** Attempt all the **05 (five)** questions.Each question carries **02 (two)** marks.The answer of each question should not exceed **50** words. $5 \times 2 = 10$

1. (a) Define communication link.
- (b) What are bit-oriented protocols?

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- (c) Define connection management.
- (d) What is error detection?
- (e) Define attenuation.

Section-B**(Short Answer Type Questions)****Note :** Attempt any **05 (five)** questions out of total **08 (eight)** questions. Each question carries **05 (five)** marks and answer of each question should not exceed **100** words. $5 \times 5 = 25$

2. (a) Explain the concept of topology and its types.
- (b) Discuss the categories of networks with example.
- (c) Explain the difference between analog and digital transmission.
- (d) Describe flow control techniques with examples.
- (e) Explain noise and its types in communication systems.

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- (f) What is multiplexing? Explain its types.
- (g) Define LCP and NCP and state their functions.
- (h) Explain transport layer protocols.

Section-C

(Long Answer Type Questions)

Note : Attempt any **02 (two)** questions out of total **04 (four)** questions. Each question carries **20 (twenty)** marks and answer of each question should not exceed **400** words. **20×2=40**

3. (a) (i) Explain OSI layers with suitable diagram and describe each layer briefly. **(10×2=20)**
- (ii) Compare OSI and TCP/IP models.

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- (b) (i) Describe the error detection and correction techniques with examples. **(10×2=20)**
- (ii) Write short notes on ARQ protocols.
- (c) (i) Explain FDM and TDM with diagrams. **(10×2=20)**
- (ii) Differentiate between message switching and packet switching. **(10×2=20)**
- (d) (i) Explain the working of repeaters, routers, and gateways with diagrams. **(10×2=20)**
- (ii) Discuss the design issues of network layer and inter-networking concepts.

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